# Does the Computer Understand? Searle's Chinese Room

*An original lesson plan inspired by **David Cole**'s the Stanford Encyclopedia of Philosophy entry on the Chinese Room Argument. Not affiliated with or endorsed by Stanford Encyclopedia of Philosophy or David Cole. Explore the original course: https://plato.stanford.edu/entries/chinese-room/*

| GRADE LEVEL | DURATION | TOPICS |
|---|---|---|
| Grades 11-12 / Intro College | 60 minutes | Philosophy of Mind |

> *"If a person inside a room can produce flawless Chinese answers by following a rulebook, without understanding a word of Chinese, does that show a computer running a program could never truly understand anything either?"*

## Learning Objectives

By the end of this lesson, students will be able to:

- Explain John Searle's Chinese Room thought experiment and the conclusion it is meant to establish about programs, syntax, and semantics.
- Distinguish the Systems Reply, the Robot Reply, and the Brain Simulator Reply, and state how Searle responds to each.
- Evaluate whether the Chinese Room argument successfully refutes 'strong AI' — the claim that a correctly programmed computer would have a mind in exactly the sense human beings do.

## Materials Needed

- Handout laying out the Chinese Room scenario step by step, with a simple diagram of the room, the slips of paper, and the rulebook
- A pre-made 'mini rulebook' activity kit: cards with invented symbols and a lookup table mapping symbol-strings to symbol-strings, with zero translation given
- Whiteboard divided into a Syntax / Semantics two-column chart
- Reply-cards handout: one paragraph each stating the Systems Reply, Robot Reply, and Brain Simulator Reply, with blank space for a rebuttal

## Lesson Procedure

### Hook: Be the Room

10 min

Before explaining anything, hand one student an envelope of cards printed with invented squiggly symbols and a rulebook that maps input-symbol-strings to output-symbol-strings purely by shape, with no meanings given. Have another 'native speaker' student slip in a question card; the first student uses only the rulebook to produce a response card, with zero idea what either card says. Ask the class whether that student understood the conversation, and hold the disagreement for later.

### Direct Instruction: The Argument and Its Target

15 min

Formally present the thought experiment: Searle imagines himself in a room with an English rulebook correlating Chinese symbols, so the notes passed in and out are indistinguishable from a native Chinese speaker's answers, yet Searle understands no Chinese. Explain that the target is 'strong AI' — the claim that running the right program is sufficient for a mind, i.e. that understanding is just formal symbol manipulation. Draw a syntax (form/shape rules) versus semantics (meaning) distinction on the board and explain Searle's diagnosis: syntax alone never gets you semantics, no matter how complex the rulebook.

### Jigsaw Group Work: Meet the Replies

20 min

Divide the class into groups, each assigned one of the Systems Reply (it's the whole system, not the man, that understands), the Robot Reply (add sensors and effectors so symbols get grounded in the world), or the Brain Simulator Reply (simulate the actual neuron-firing patterns of a Chinese speaker). Each group must state its assigned reply in its own words and then construct Searle's actual rebuttal to it — for example, that memorizing the whole rulebook makes the man himself the entire system, and he still understands nothing.

### Whole-Class Synthesis Debate

10 min

Reconvene and have each group present its reply and Searle's rebuttal in about two minutes, then open the floor: does any reply actually escape Searle's rebuttal? Push the discussion toward whether 'understanding' requires a biological substrate or could be substrate-independent.

### Closure: Exit Ticket

5 min

Students individually write one sentence answering: what is the single most important difference between syntax and semantics, and why does Searle think it matters for AI? Collect these as a quick formative check of whether the core distinction landed.

## Discussion Questions

1. Is understanding simply very sophisticated symbol manipulation, or does Searle's thought experiment show it must be something more?
2. Does the Systems Reply succeed by pointing out that the man is just one part of a larger system, or is it merely relocating the mystery rather than solving it?
3. If a robot's symbols were grounded in cameras and sensors connected to the real world, would that change whether it understands, or is Searle right that it's still just more meaningless input?

4. Searle argues understanding requires the right biological 'causal powers' of the brain. Is this a real argument, or just an assertion that only brains can think?
5. Could Searle's own argument be turned against human neurons — after all, no single neuron understands Chinese either?

## Assessment

Students write a two-paragraph response in which they state, in their own words, what conclusion Searle believes the Chinese Room establishes about programs and understanding, and then pick one reply (Systems, Robot, or Brain Simulator) and argue whether it successfully defeats Searle's argument or ultimately fails, citing a specific feature of the thought experiment to support their verdict. Grade for accurate representation of the argument and for direct engagement with the chosen objection rather than a dismissal of it.

## Extension Activity

Have students find a transcript or example of a conversation with a modern AI chatbot and write a short analysis applying the Chinese Room argument directly to it: does the fact that the chatbot's replies look fluent and contextually apt settle whether it understands anything, according to Searle's reasoning? Students should identify what a defender of strong AI might say in response.